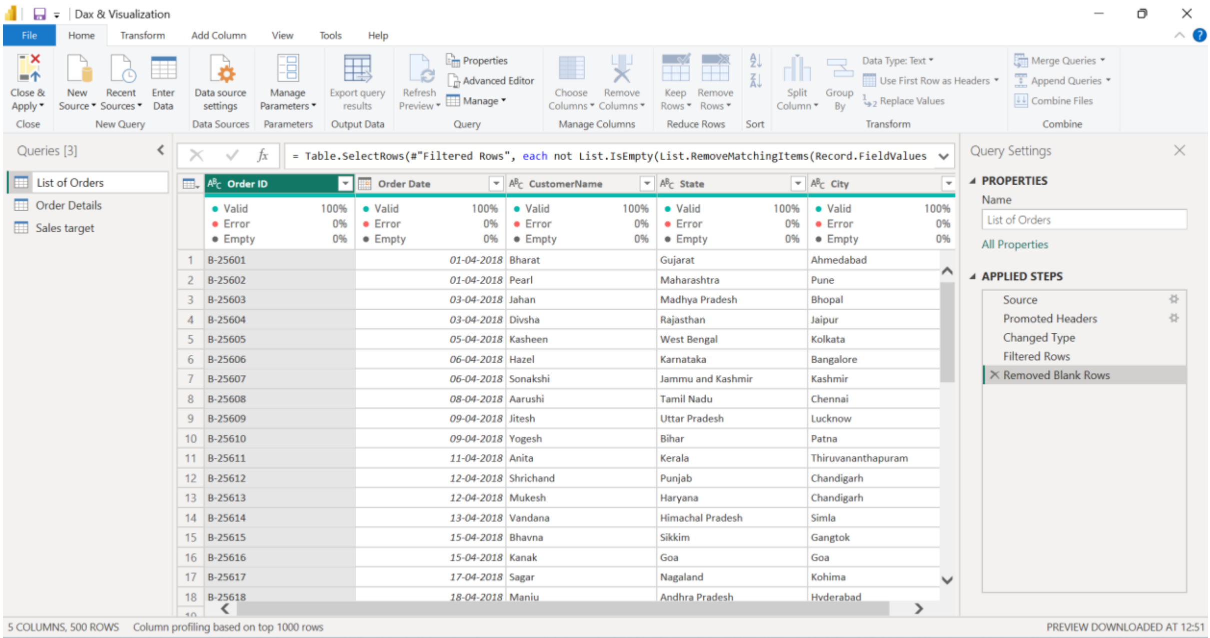
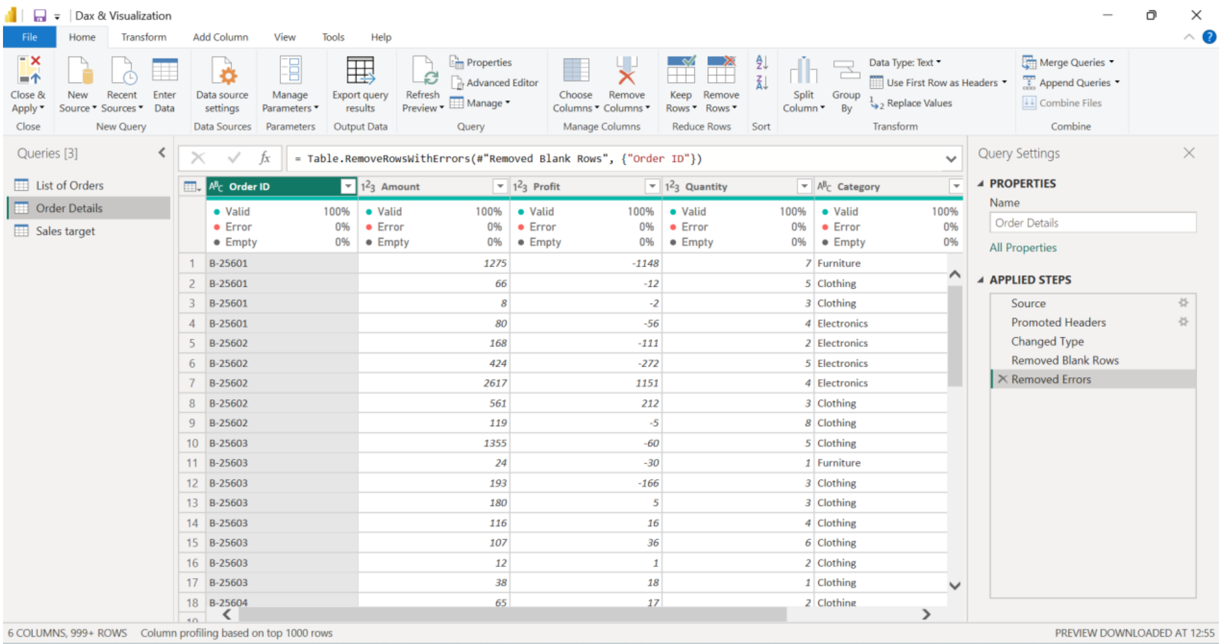
Power BI Assignment 2 - DAX & Data Visualization

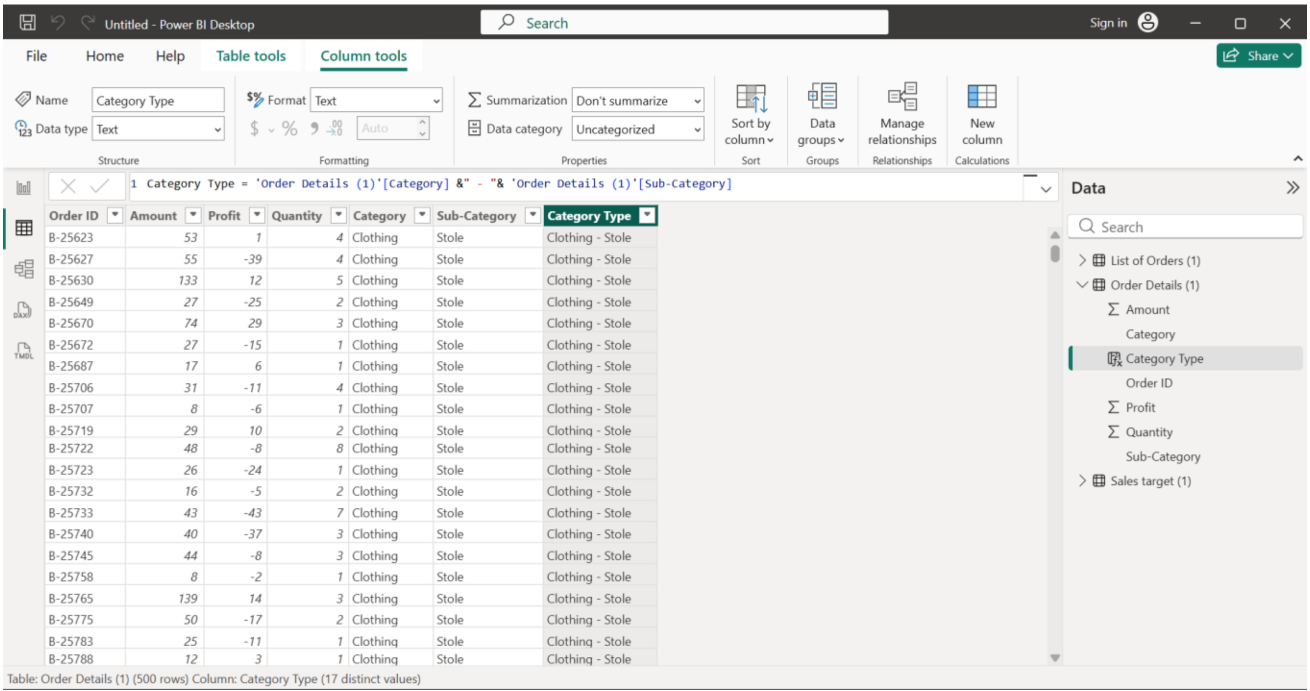
E-Commerce Sales Analysis:

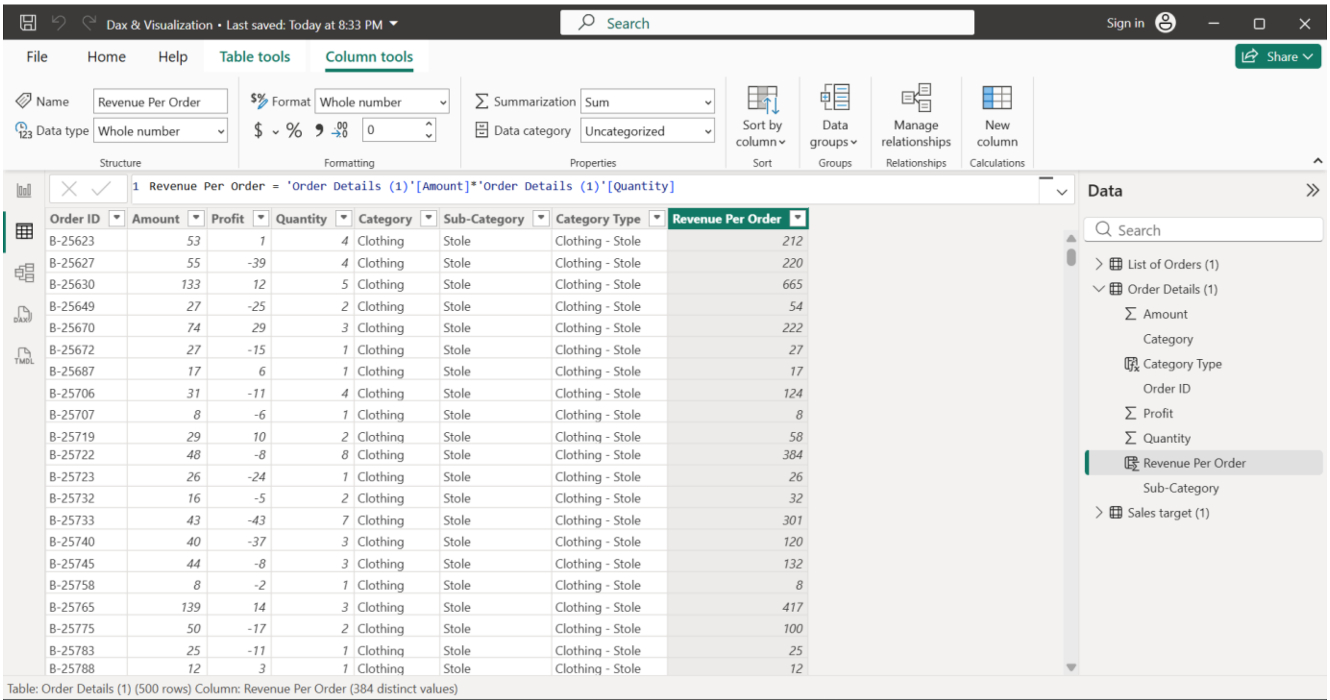
In This Assignment, I Analyzed E-commerce sales data using Power BI and DAX. Using Datasets List of orders, Order details,Sales Target. I imported and connected multiple datasets, created calculated columns and measures such as revenue, order count, and YTD sales, and developed visualizations to analyze sales trend, targets, profit, quantity, and regional performance. This project helped me improve my skills in Data modeling, DAX Formulas, KPI, charts and dashboard Visualization.

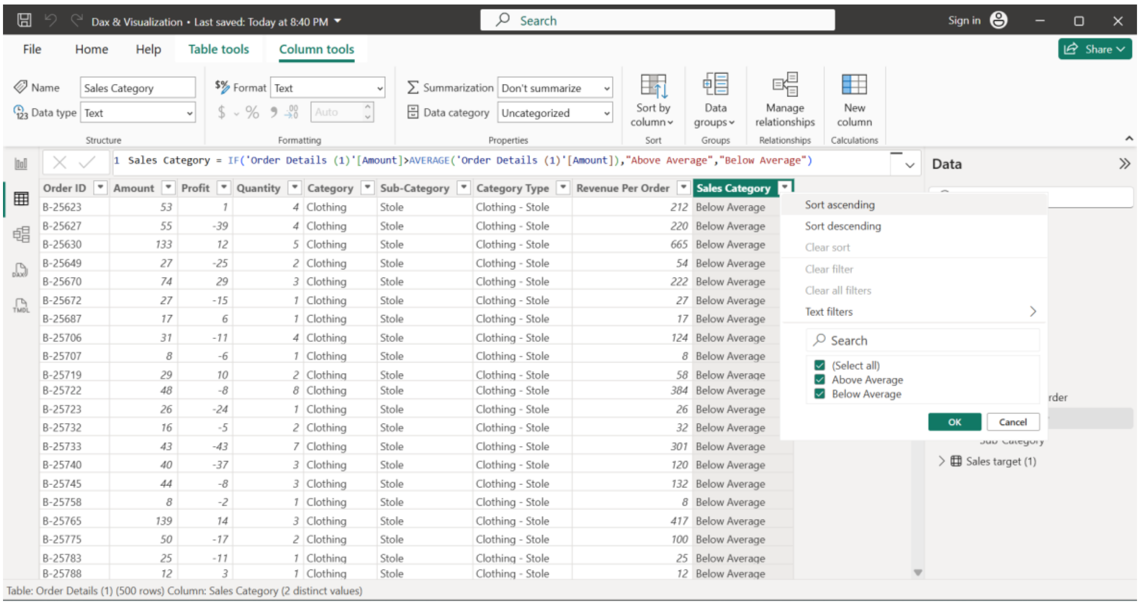


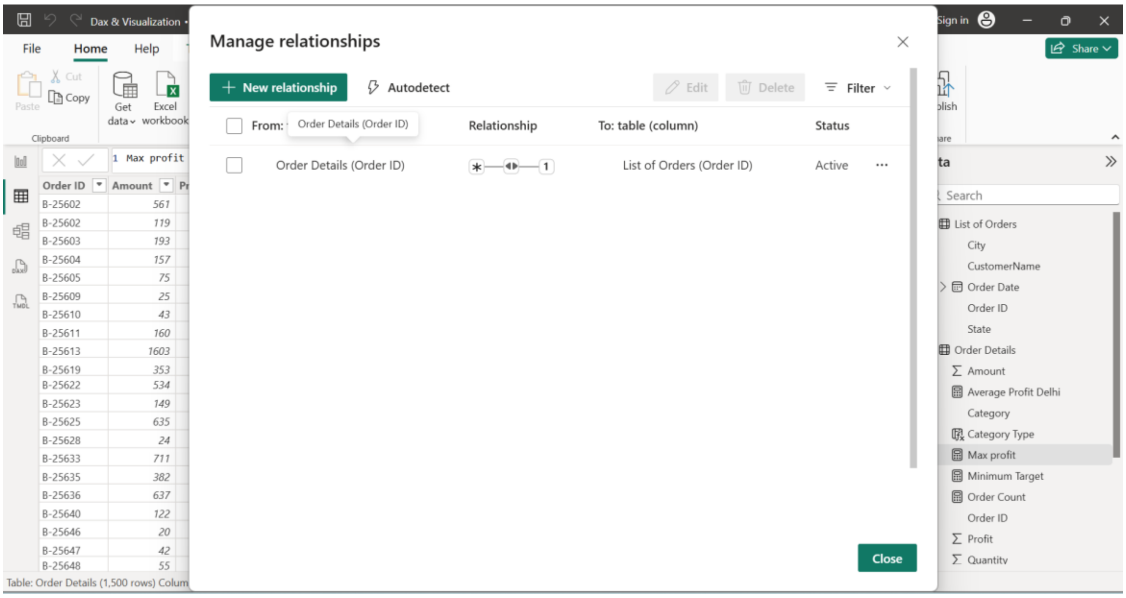


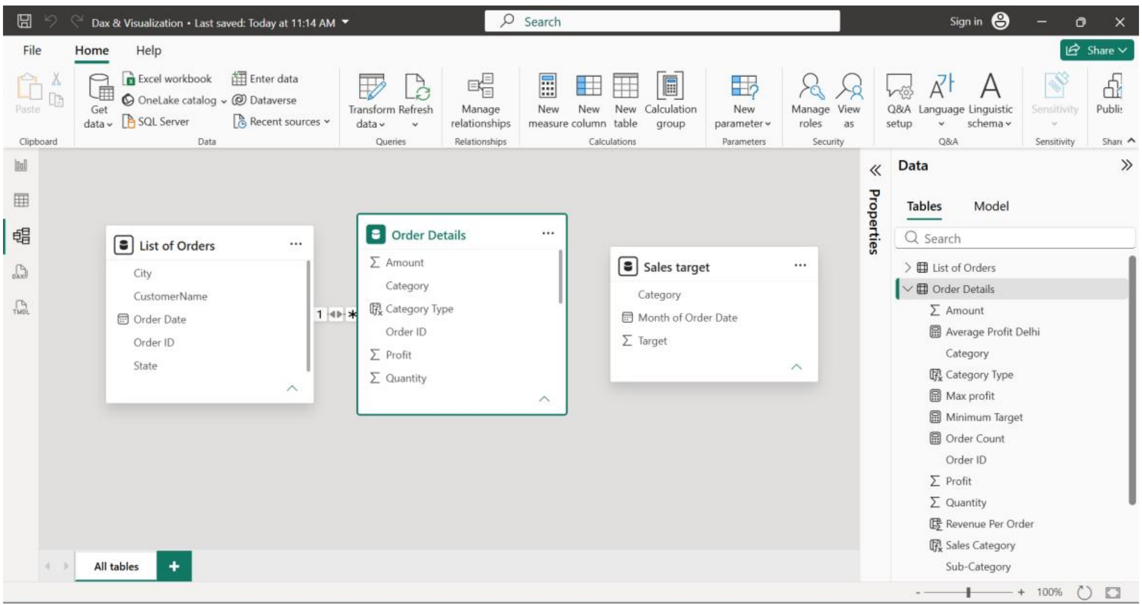
Calculated Columns:



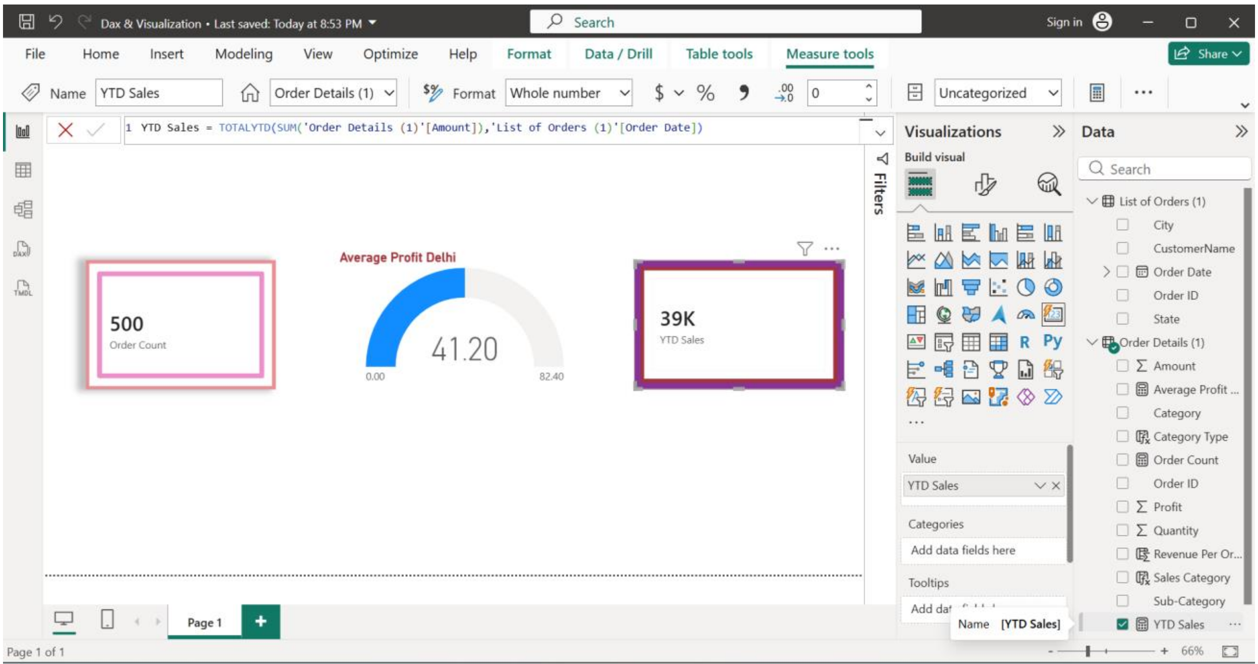



Dax & Visualization • Last saved: Today at 8:40 PM
File Home Help Table tools Column tools
Name Sales Category
Data type Text
Format Text
Summarization Don't summarize
Data category Uncategorized
1 Sales Category = IF('Order Details (1)'[Amount]>AVERAGE('Order Details (1)'[Amount]),"Above Average","Below Average")
Order ID Amount Profit Quantity Category Sub-Category Category Type Revenue Per Order Sales Category
Sort ascending
Sort descending
Clear sort
Clear filter
Clear all filters
Text filters
(Select all)
Above Average
Below Average
OK Cancel
Table: Order Details (1) (500 rows) Column: Sales Category (2 distinct values)


Manage relationships
New relationship
Autodetect
Edit Delete Filter
From: table (column) Relationship To: table (column) Status
Order Details (Order ID) List of Orders (Order ID) Active
Close

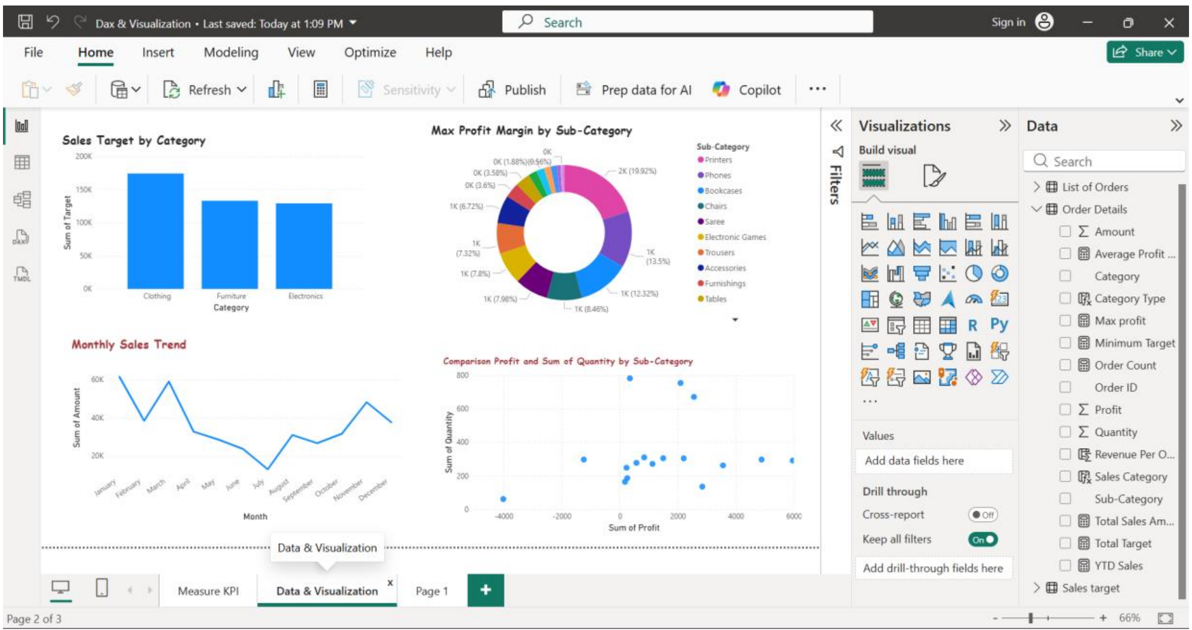

Dax & Visualization • Last saved: Today at 11:14 AM
List of Orders
City
CustomerName
Order Date
Order ID
State
Order Details
Amount
Category
Category Type
Order ID
Profit
Quantity
Sales target
Category
Month of Order Date
Target
All tables

Calculated Measures / KPI:



- Order Count = Count (‘Order Detail’ [Order ID])
- Average Profit in Delhi = CALCULATE(Average(‘Order Details’[Profit]),’List of Orders’[City]= “Delhi”
- YTD Sales = TOTALYTD(SUM(‘Order Details’[Amount],’List of Orders’[Order Date])

Data Visualization:



1. Sales Target Achievement by Category

   Using a Clustered Column Chart : X Axis →Category, Y Axis →Sales Target

2. Max Profit Margin by Sub-Category

   Using a Donut Chart : Legend →Sub-Category, Values →Max Profit

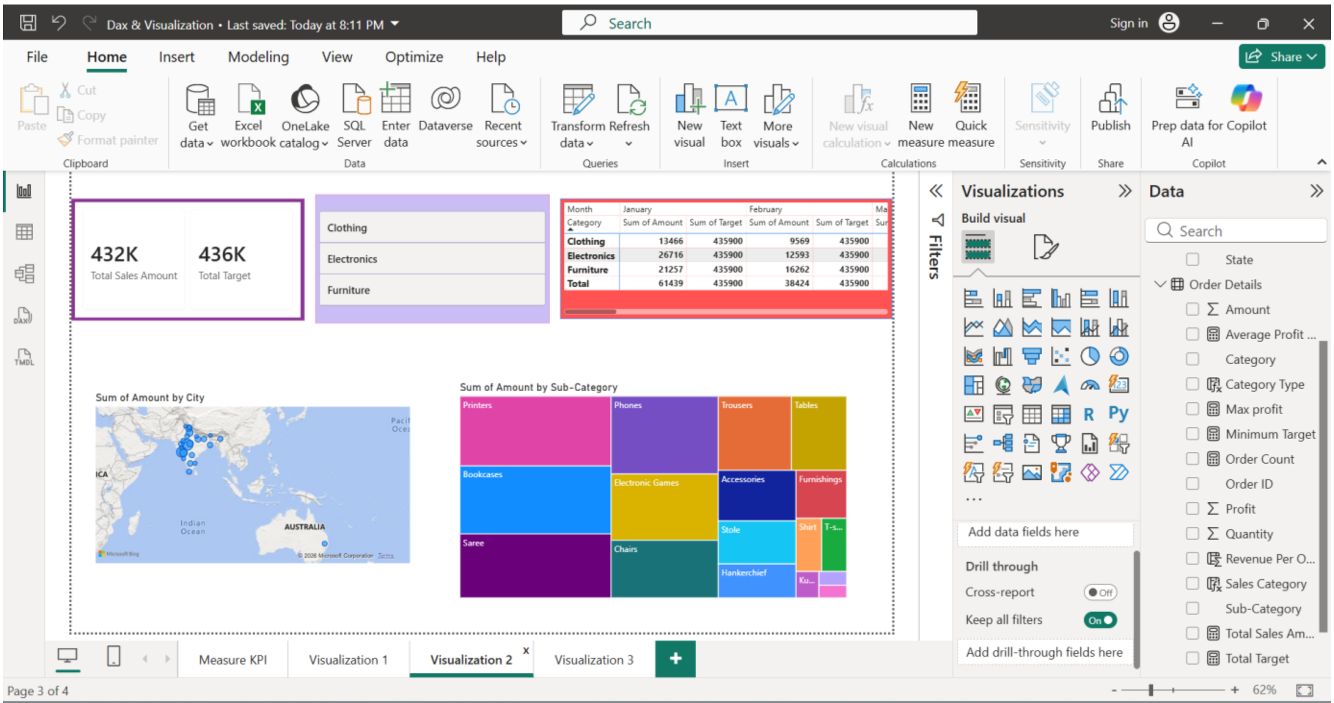
3. Monthly Sales Trend

   Using a Line Chart : X Axis →Order Date(Month), Y Axis →Sum of Amount

4. Comparison of Profit and Quality by Sub-Category:

   Using a Scatter Chart: X Axis→Sum of Quantity, Y Axis →Sum of Profit

   Values →Sub-Category



5. Comparison of Total Sales Amount and Target: Using Card Visual

6. Sales Performance Matrix: Row →Category, Columns→Order Date(Month)

   Values→Sum of Amount, Sum of Target

7. Geographic Sales Analysis: Using Visualize Map

8. Sales Distribution by Sub Category: Using a Treemap Visual

Category →Sub-Category, Values→ Sum of Amount

9. Order Count Analysis by State: Create a Funnel Chart

Category→State, Values→Order Count